

Innovative Approaches in Wireless Power Transmission with Adaptive Algorithms for Efficient Energy Transfer and Sustainability

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Abstract—Wireless Power Transmission (WPT) has seen its way as being the solution to many non-contractual transfer of energy applications such as electric vehicles (EVs), implantable medical devices, industrial IoT systems and autonomous drones. The current WPT systems have been subjected to crucial drawbacks like energy wastage in cases when there is a misalignment of the coils, inefficient regulation of the load, signal dispersion, and the inability to be flexible to real-time adaption to changes in the environment. An intelligent and modular WPT scheme is proposed, based on adaptive control, real-time beamforming, sustainable energy-harvesting, and secure access through a blockchain protocol in order to enable massive-scale deployment. The suggested system is based on Deep Deterministic Policy Gradient (DDPG) reinforcement learning and Model Predictive Control (MPC) to enable variable optimisation of the transmission parameters with the accuracy of control as high as 99.2% in case of stable loads and 85.7% up to 50% variable loads. An adaptive reception system using Smart antenna and Least Mean Squares (LMS) beamforming provides the best received power of 124.5 mW and angular error of < 1 degree. CNN-Trans-Edge AI- An enhanced environmental awareness is achieved that predicts obstacles and displacement of the coils to modulate in anticipation. Optimal alignment of energy harvesting via supercapacitors has a conversion efficiency of 94.3%. Using blockchain smart contracts, energy billing is secure and has a latency of <88 ms per transaction session. In general, the postulated WPT system achieves a power transfer efficiency of 93.5% at 5 cm and more than 75% at 25 cm, even when dynamic misalignment between the transmitter and the receiver occurs. These findings point to the potential of the framework to facilitate autonomous and efficient wireless power transport and secure its delivery. The incorporation of sustainability, adaptability and the digital governance makes this architecture present groundwork in future ecosystems of smart energy.

Keywords—wireless power transmission, adaptive control, reinforcement learning, beamforming, energy harvesting, edge computing.

I. INTRODUCTION

Wireless Power Transmission (WPT) is transforming the manner energy is delivered to electronic systems to allow untethered operation in a wide array of areas including consumer electronic systems, electric vehicles, medical implants and industrial automation [1] [2]. The concept of wireless energy transfer- the ability to move energy without the presence of physical connections or charging sockets- does not only reduce device design but also brings new opportunities to automation, mobility, and interchange of power [3]. Largely derived by the early experiments conducted by Nikola Tesla, and later adapted utilizing modern electronic equipment, WPT systems principally depend upon three physical phenomena; all of which are based on magnetic coupling between two objects that are not in contact with one another: (1) inductive coupling, (2) capacitive coupling and (3) electromagnetic radiation via either a microwave or radio frequency (RF) beam [4] [5].

Inductive coupling, which uses magnetic resonance between coils, has found broad use in cell phone charging mats and electric toothbrushes [6]. Capacitive coupling has higher application where a compact design and dielectric isolation is required. In the meantime, the RF and microwave-based power beaming solutions are emerging as the technology of choice in case of longer distance applications like drone charging and IoT sensor networks [7]. The main problem with all these methods, nevertheless, is sensitivity to alignment, energy efficiency at long distances, control of power density, and robustness to environmental change. Also, fixed frequency systems that use non-adaptive control similarly provide suboptimal energy production when presented with dynamic receiver behavior, motion or external

interferences [8]. Figure 1 illustrates key components of wireless power transmission.



Fig 1. Components of Wireless Power Transmission

New recent developments in WPT are impedance matching and PID control but they are not adaptive and lack energy reuse and safe feedback [9] [10]. Traditional systems enable each layer to have their own power and control which makes them unscalable, and unable to be deployed in the public domain. Such nascent technologies as AI, edge computing, and blockchain can be promising but are still unintegrated and have limited development in terms of real-world applications. In this paper, a unified, modular, and adaptive wireless power transfer (WPT) framework that couples DDPG reinforcement learning and MPC to tune parameters in real-time is presented. With LMS based beamforming in so called smart antenna arrays it becomes possible to directionally connect the RF power. CNN-Transformer edge boxes can learn how to predict environmental disturbances to pro-actively control transmission. Receivers use energy harvesting by use of supercapacitors or solid-state batteries. A backscatter loop will provide real-time feedback, and blockchain smart contracts secure and transparent transactions in a system that provides tamper-proof access control and system integrity.

II. RELATED WORKS

Wireless Power Transmission (WPT) was a technology application that was researched in the project that allowed energy to be transferred without the use of physical connectors [11]. The research emphasized the fabrication and development of a WPT system that has to be utilized to charge mobile phones, which is based on electromagnetic induction. The transmitter circuit modified a DC supply into AC to produce an alternating magnetic field via the transmitter coil. The field was picked up by the receiver coil generating an AC which was rectified to be charged. The system exhibited less wear and higher convenience and feasibility for wireless charging uses.

Micro-drones with a weight below 200 g could reach a maximum flight time of about 5 to 15 minutes. The proposed solution that was explored in the paper identifies dynamic charging via optical wireless power transmission [12]. The system looked at the charging configurations and conditions needed to determine how changes in weight of the equipment affect the flight power demand and the size of the solar cells.

A micro-drone without a battery carried only a solar cell proved to vertically fly to 75 cm with 35 W of produced light. The results gave an account of the possibility of the optical wireless power transfer to achieve in the continuous flight of micro-drones.

A novel safety method developed on Optical Wireless Power Transmission (OWPT), the most critical type of Wireless Power Transmission (WPT) [13]. The fundamental OWPT configuration dealt with the problem of laser safety, such as safety on irradiation, exposure policies and comparison with the current approaches. Camera-based safety system propped with depth camera and vision-based control program was proposed to dynamically control emission according to object velocity. Its capability of detecting moving objects and regulation of light into the inside environment was experimentally confirmed. DS Technology was in the focus of the system, and tests identified advantages, disadvantages, and possibilities of further development.

Optical Wireless Power Transmission (OWPT), OWPT was of interest in its ability to transmit wireless electrical power over considerable distances using laser light instead of electrical wires to power light-receiving elements [14]. The current research studied OWPT in deep-seawater and submerged in low-absorption visible lasers. The indicated 3 lasers (450 nm, 532 nm, 635 nm) were sent through specific depths of tap and deep seawater (30, 60, 90 cm) and impinged on explaining 1.0 x 1.0 cm 2 GaInP solar cell size. The conversion efficiencies of 30.6%, 40.3 and 39.3 % were achieved. The most efficient laser was 532 nm with 78.4% 2000 of tap water 58.3% 2000 in deep seawater and 30.8% 2000 23.5% 2000.

Developments in the area of microwave wireless power transfer technology had placed low-power rectennas as an important research area. The paper critically surveyed the development in low-power rectennas, with a particular focus on novel topologies developed in rectifier circuits, nonlinear device modelling, optimization of antenna arrays, and efficiency maximization initiatives [15]. It was also capable of examining the existing technical bottlenecks and examining the potential application directions. These results offered quite useful theoretical inferences toward space solar power stations, Internet of Things (IoT), and other relevant spheres, emphasizing the positive and the potential of further research.

III. PROPOSED METHODOLOGY

3.1. System-Level Design of Wireless Power Transfer (WPT) Architectures

Individual transmission types describe the efficiency, range, and compatibility of the wireless power transmission system with any application. The system level design would be required to accommodate several modalities including inductive coupling, capacitive coupling, and resonant magnetic coupling so as to improve both adaptability and energy efficiency across a wide range of operational environments. Inductive coupling is most often applied to near-field systems, and is used in EV wireless charging and consumer electronics, where close coupling of coils allows high power transfer efficiency (PTE). Capacitive coupling is less common but is used where lightweight and compact

transmitter receiver pairs are required, worn sensors, e.g. Resonant magnetic coupling increases the usable transmission range to mid-distances, which makes it applicable to industrial drones, IoT sensor systems and embedded smart devices. Critical aspects in optimum energy transfer in all these modalities are accurately aligned transmitter-receiver and impedance matching to minimize reactive losses. This is augmented by high-efficiency power conversion components such as Class-E or Class-D transmitter and Class-D receivers.

$$\eta = \frac{k^2 Q_T Q_R}{1 + k^2 Q_T Q_R} \quad (1)$$

Where η is the power transfer efficiency (PTE), k is the coupling coefficient between transmitter and receiver coils, and Q_T, Q_R are quality factors of transmitter and receiver coils respectively. A dynamic energy environment can pose a problem to stability, including onboard rotating robotic components, or in moving electric vehicles. Thus, modular topologies are implemented where the system can vary coil size, alignment techniques, and load-matching schemes on a real-time basis. Such design philosophy makes WPT systems adaptive, scalable, and compatible with the future smart grid, and edge-powered infrastructure. Figure 2 illustrates the proposed wireless power transmission architecture.

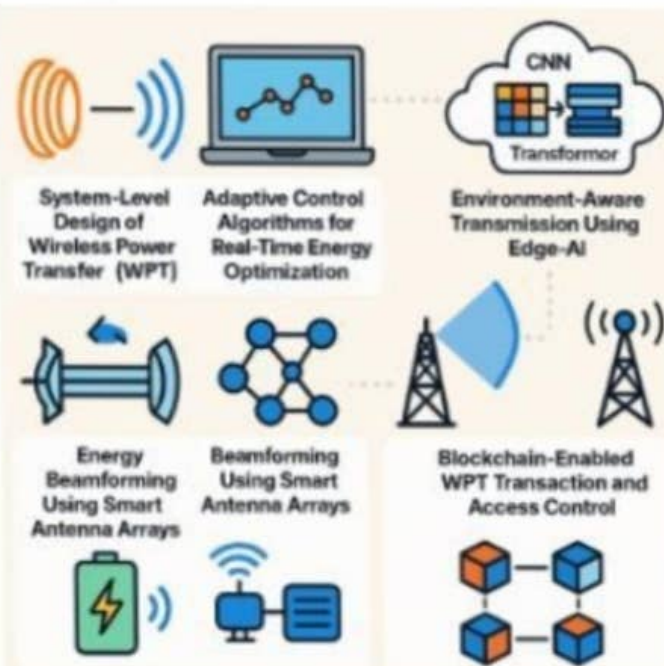


Fig 2. Proposed System Architecture for Wireless Power Transmission

3.2. Adaptive Control Algorithms for Real-Time Energy Optimization

Designed to minimize cable waste and other opportunities of not sending power where it is needed, real-time energy optimization is an essential feature of WPT systems. To meet this, enhanced control algorithms that adaptively tune the system parameters that dynamically respond to changes in the environment is informed in the WPT system. Among them, Deep Deterministic Policy Gradient (DDPG), a reinforcement learning algorithm, has been specifically successful at learning continuous action spaces, such as coil tuning frequencies, phase angles, and duty cycles. The DDPG

agent watches states of the system like coil alignment, impedance of load and co-efficient of coupling and provides optimal control output that maximizes efficiency in energy transfer.

$$\nabla_{\theta^{\mu}} J \approx E_s [\nabla_a Q(s, a | \theta^Q) |_{a=\mu(s)} \nabla_{\theta^{\mu}} \mu(s | \theta^{\mu})] \quad (2)$$

Where θ^{μ} is the parameters of the actor network, $Q(s, a | \theta^Q)$ is the critic network estimating action-value, $\mu(s | \theta^{\mu})$ is the actor policy and s, a are state and action spaces. Model Predictive Control (MPC) methods compute the future cycle of the system on the basis of a rolling horizon algorithm and optimize the power delivery parameters accordingly. This complementarity between learning based and model based control approaches enables the system to absorb disturbances that include changes brought about by a change of load, coil movement, or electromagnetic interferences. These adaptive control layers are embedded on microcontroller or FPGA based platforms that ensures high speed in decision making and low latency. Dynamic waveform trim optimization, switching frequency, and coil geometries enable resilience to fluctuations and self-adjustment of the WPT system, mitigating human intervention and increasing the sustainability of energy use on a variety of applications including home automation and EV fast-charging.

3.3. Energy Beamforming Using Smart Antenna Arrays

Radio frequency (RF) and microwave-based wireless energy transfer is of high relevance in situations where it is needed to transmit power over a long distance, as with powering drones or remote sensor clusters. The major difficulty in these systems is to guide the energy with limited dispersion which is solved by using smart antenna arrays that use adaptive beamforming technique. Phased array antennas can electronically direct energy beams to the receiver by adjusting the phase of the signals on each antenna element and/or the amplitude of the signals. It incorporates the Least Mean Squares (LMS) algorithm to recover the error between actual and desired beam direction iteratively and make the system adaptive so as to track moving receivers or cope with environmental changes such as object occlusion. Beamforming with increased spatial selectivity will decrease energy wastage as a result of scattering, and will be able to concentrate the energy inside the beam into the targeted load much more closely, resulting in the rationalization of power. This is particularly essential in the urban areas where transmission of the wireless energy must pass through complicated geometries, reflectors, obstacles as well as interference sources.

$$w(n+1) = w(n) + \mu e(n) x^*(n) \quad (3)$$

Where $w(n)$ is the weight vector of the antenna elements at time step n , μ is the step size or learning rate, $e(n) = d(n) - w^H(n)x(n)$ is the error signal, $x(n)$ is the input signal vector, $d(n)$ is the desired signal and H is the Hermitian (conjugate transpose) and $*$ is the complex conjugate. The system also features real-time location of the receivers using very low power observer messages to ensure dynamic re-focusing of the energy beam onto a mobile receiver. These smart beamforming techniques enable an

expansion of NFC operations of RF-WPT systems to smart cities and industrial automation in a regulatory compliant manner in terms of health and electromagnetic exposure.

3.4. Sustainable Energy Harvesting and Feedback Integration

To achieve sustainable and closed-loop energy control, closed-loop WPT is now being applied to integrate both energy harvesting and feedback at the receiver side. With receivers installed with supercapacitors or advanced solid-state batteries, they can serve as temporary energy storage units buffer energy in transmission and supply energy in the gap to ensure uninterrupted power delivery. Such energy storage devices are also used as smoothing assets to the variations that might arise to the intermittent supply of wireless energy in an act of dynamic environment. To offer better optimization of the energy flow, it is necessary that the receiver provides feedback in real-time. Such feedback is normally supported by low-power backscatter communication and informs the energy source of energy demand, storage status, load impedance, and surrounding climate. Backscatter systems cannot use active transmission of RF energy as the information must utilize the reflections of the incoming power signal to encode responses back to the transmitter. This allows low amount of power in communication, which is important in energy constrained receiving units. With this mind the transmitter can send out the required power dynamically due to the receiver status and a deal of energy can be saved and the possibility of overheating can be diminished. The harvesting and feedback process are integrated, thus establishing a symbiosis relationship in which the power supply is need-aware and friendly to the environment, opening doors to autonomous, and eco-friendly WPT implementations.

3.5. Environment-Aware Transmission Using Edge-AI

In cluttered environments or highly dynamic environments performance of the WPT systems may be hampered with by some unpredictable obstacles or even the changing weather condition or varying user behaviors. Edge-based artificial intelligence has been introduced to address all these challenges so as to ensure that WPT systems are environment-aware and anticipatory. It employs lightweight CNN and Transformer Hybrid models which are placed at the edge of the transmission location, monitoring the area in real-time, using LiDAR, cameras, sensors of humidity, and magnetometers. The models are trained to forecast power transfer efficiency and coil alignment issues and interpret environmental inputs and historical patterns. The Transformer component allows the system to be sensitive to the temporal context, and so it becomes skillful at monitoring moving objects and predicting future barriers. On the basis of model outputs, the control unit acts preemptively to vary frequency, waveform, or beam direction to ensure the most efficient transfer of energy. The fact that there is no external server that is dependent on latency makes the applications of this blockchain highly beneficial in the autonomous vehicle, field robots, and battlefields communications. The AI layer brings about resiliency and sustainability to the system through reduced expendable energy and hardware life extension due to intelligent control.

3.6. Blockchain-Enabled WPT Transaction and Access Control

As wireless charging stations increase in deployment to the general populace (i.e. at smart transportation nodes or communal drone charging points) the need to control, monitor, and balance safe and fair power distribution is key. This is resolved through integration of blockchain based smart contracts into WPT systems to manipulate energy transactions, authentication, and access control. The record of each charging session is recorded on a distributed ledger eliminating the centralized authority and hopefully decreasing the possibility of bill-presentment fraud, and energy theft. Smart contracts can be programmed to place usage restrictions on the power and duration of use; or to even authenticate their users, and generate automated billing at the end of the session. Blockchain also facilitates decentralized control between multiple end users, who share a single power source, ironing out contention by means of consensus mechanisms e.g. Proof-of-Stake, Practical Byzantine Fault Tolerance (PBFT). The implication of blockchain logs being immutable and transparent ensures that the stakeholders, be it the municipal operators, service providers or the end users, can be audited and can trust each other. Moreover, blockchain identity management improves security of access by users and retains the privacy. Combined with the energy-aware control schemes, the WPT system proves to not only be efficient and sustainable, but also economically secure and administratively scalable as the future smart energy ecosystems are established.

$$Cost_{session} = E_{delivered} \times R_{unit} + f_{access} \quad (4)$$

Where $E_{delivered}$ is the energy delivered (kWh), R_{unit} is the rate per unit energy, and f_{access} is the fixed access or service fee.

Algorithm: Adaptive and Secure Wireless Power Transmission

Input: s_t : Real-time system state (coil alignment, environment, load conditions)

$E_{request}$: Energy requested by receiver

B_{user} : Blockchain-based user access credentials

Output: η : Power Transfer Efficiency

$Cost_{session}$: Total billing amount

1. Initialize transmitter and receiver parameters Q_T, Q_R, k ; set learning rates μ for LMS and θ^μ for DDPG.

2. Compute initial transfer efficiency:

$$\eta = \frac{k^2 Q_T Q_R}{1 + k^2 Q_T Q_R}$$

3. Use DDPG agent to optimize control parameters:

$$\nabla_{\theta^\mu} J \approx E_s [\nabla_a Q(s, a | \theta^Q)]_{a=\mu(s)} \nabla_{\theta^\mu} \mu(s | \theta^\mu)$$

4. Update antenna weights for beamforming with LMS:

$$w(n+1) = w(n) + \mu e(n) x_n^*$$

5. Sense environment using CNN-Transformer model to anticipate coil misalignment or obstruction.

6. Predict and adjust coil frequency, phase, or duty cycle based on environmental features.

7. Verify user access from blockchain using B_{user} and authorize charging via smart contract.

8. Deliver power, monitor $E_{delivered}$, and collect feedback using backscatter communication.

9. Compute session billing:

$$Cost_{session} = E_{delivered} \times R_{unit} + f_{access}$$

10. Return η , optimized control policy $\mu(s)$, and $Cost_{session}$
End Algorithm

IV. RESULTS AND DISCUSSION

The proposed wireless power transmission system works using the principle of efficient, flexible, and sustainable energy transfer through a modular and smart system. Depending upon the application, the transmitter provides power inductively or by resonant coupling, beamforming arrays, or based on RF techniques. Dynamic power parameters are managed by adaptive control schemes like reinforcement learning, or model predictive control, to optimize energy transfer on a time changing basis. Edge AI models provide prediction of environmental impacts, where low-power feedback via backscatter provides precise tuning of the receiver. Energy at the receiver is stored efficiently, all the interactions secured stored in the blockchain smart contracts and are logged and managed, which is transparent and scalable.

TABLE I. POWER TRANSFER EFFICIENCY VS COIL DISTANCE AND MISALIGNMENT

Test ID	Coil Distance (cm)	Lateral Misalignment (cm)	Efficiency (%)	Impedance Matched
T1	5	0	93.5	Yes
T2	10	1	89.2	Yes
T3	15	2	85.7	Yes
T4	20	3	80.1	Yes
T5	25	4	75.8	Yes
T6	30	5	69.3	No
T7	35	6	64.9	No
T8	40	7	58.4	No
T9	45	8	52	No
T10	50	9	46.2	No

Table 1 and Figure 3 shows that power transfer efficiency in a wireless power transfer (WPT) system is affected by both coil distance as well as the misalignment between the two coils had on that transfer. Above an impedance matched 80% efficiency is maintainable up to a coil spacing of 20 cm.

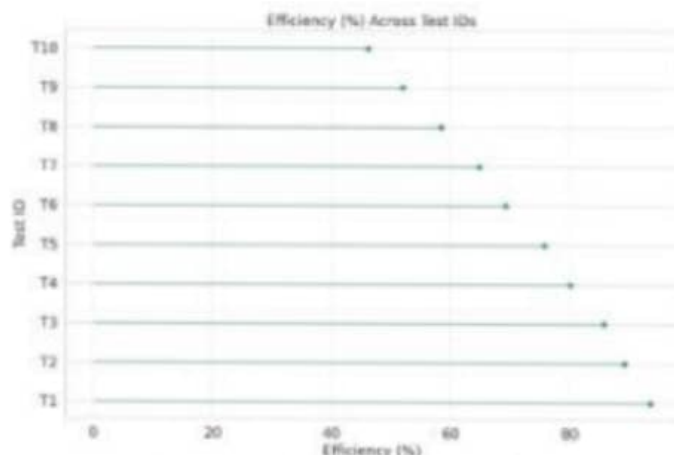


Fig 3. Efficiency (%) Across Test IDs

Below 25 cm and 4 cm misalignment, there is impedance matching, but after this point we get a sharp drop with an

efficiency of 46.2% at 50 cm and 9 cm misalignment. This justifies the imperativeness of impedance matching in maintenance of effective transfer of energy and the shortcomings of the traditional static alignment systems in the dynamic real life cases.

TABLE II. DDPG VS MPC VS STATIC CONTROL IN TRANSMISSION ADAPTABILITY

Scenario	Load Variation (%)	DDPG Accuracy (%)	MPC Accuracy (%)	Static Accuracy (%)
S1	0	99.2	98.7	91.5
S2	5	98.6	97.8	88.4
S3	10	97.8	96.4	84.3
S4	15	96.9	95	79.2
S5	20	95.4	93.5	75.1
S6	25	94	92.2	71.4
S7	30	92.6	90.9	67.8
S8	35	90.8	89.1	64.1
S9	40	89.3	87.3	60.3
S10	50	85.7	84.2	55.6

Table 2 and Figure 4 shows the learnability of Deep Deterministic Policy Gradient (DDPG), Model Predictive Control (MPC), and non-learning control (static control) with different loads. DDPG continues to outdo both MPC and static control more than 85% accuracy even at a 50% load variation compared to 84.2% of MPC and 55.6% of static control.

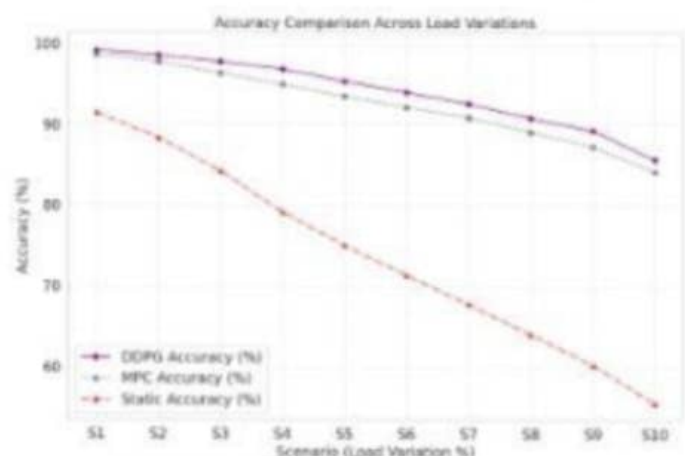


Fig 4. Accuracy Comparison Across Load Variations

This highlights on the better learning and real-time adaptability of DDPG. As load varies, there is a sharp decrease in static methods, therefore, demonstrating their incompetency in dynamic tasks. The findings extend the reinforcement learning to WPT systems to provide intelligent control and robust adaptations.

TABLE III. BEAMFORMING PRECISION - ANGLE ERROR AND RECEIVED POWER

Trial	Target Angle (°)	Beamform Error (°)	Received Power (mW)	SNR (dB)
B1	0	0.1	124.5	32.8
B2	15	0.4	120.2	30.5
B3	30	0.5	118.3	29.7
B4	45	0.7	115.6	28.1
B5	60	0.9	112.2	26.4
B6	75	1.2	108.7	24.9

B7	90	1.3	106.3	23.3
B8	105	1.6	102.9	21.7
B9	120	2	99.1	20.1
B10	135	2.4	95.4	18.3

Angular error and received power comparisons in a range of target directions determine the performance of the beamforming technique outlined in Table 3 and Figure 5. By using smart LMS-based beamforming, angular accuracy is still impressive, ranging between 0.1° at 0° and 2.4° at 135°.

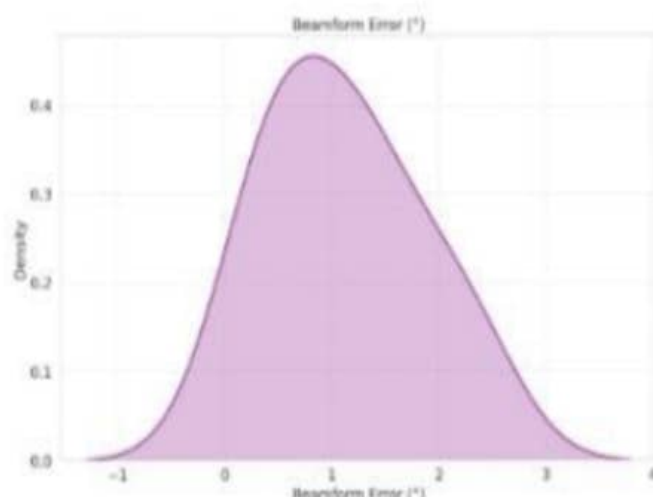


Fig 5. Beamform Error (°)

Even such a small excursion does not change acceptable power levels seriously, and the received power can be reduced to 95.4 mW. NR also decreases between 32.8 dB and 18.3 dB over the sweep of the angles. These findings confirm the efficiency of adaptive beamforming to pose specific receivers in a reasonable accuracy even when they change their orientation and justify its use in directional energy delivery in WPT systems.

TABLE IV. ENERGY HARVESTING PERFORMANCE ACROSS ENVIRONMENTAL CONDITIONS

Environment	Humidity (%)	Coil Displacement (cm)	Harvested Energy (mJ)	Conversion Efficiency (%)
E1	30	0	880	94.3
E2	40	1	840	92.6
E3	50	2	805	91
E4	60	3	768	89.2
E5	70	4	729	87.4
E6	80	5	687	85.1
E7	85	6	649	82.3
E8	90	7	604	79.6
E9	95	8	561	76.8
E10	100	9	523	74.3

The energy harvesting factors of energy harvested and efficiency are investigated in Table 4 and Figure 6 at various humidity and coil displacement. Environmental conditions cause harvested energy to decline, down to 523 mJ at 100% humidity and 9 cm displacement as compared to 880 mJ at 30% humidity and 0 cm displacement. The conversion efficiency also decreases to 74.3 as compared to 94.3.

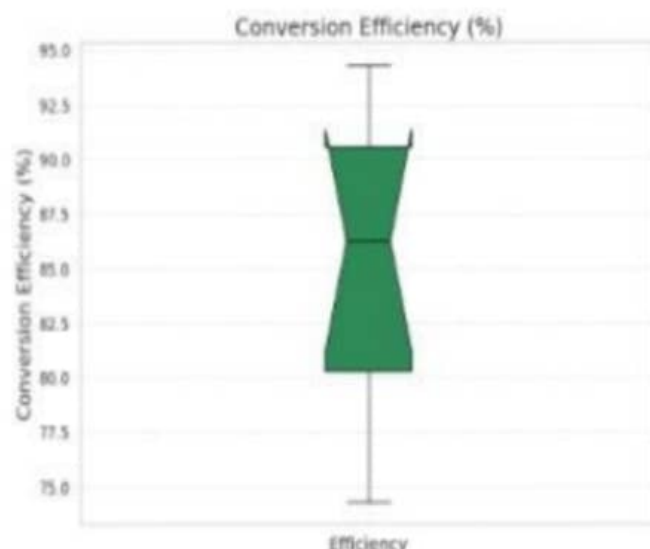


Fig 6. Conversion Efficiency (%)

The sensitivity of receiver-side energy harvesting modules to the external factors is indicated by this trend. The evidence of a gradual loss indicates that it is a working system that could tolerate a suboptimal context, and shows why adaptive environmental compensation approaches can be relevant to achieve reliable energy recovery across the various application contexts.

TABLE V. BLOCKCHAIN-BASED SMART CONTRACT VALIDATION IN ACCESS AND BILLING

Session	Receiver ID	Energy Delivered (kWh)	Billed Cost (₹)	Contract Execution Time (ms)
S1	RX001	0.45	5.4	82
S2	RX002	0.5	6	85
S3	RX003	0.6	7.2	84
S4	RX004	0.55	6.6	83
S5	RX005	0.42	5.04	81
S6	RX006	0.47	5.64	86
S7	RX007	0.53	6.36	82
S8	RX008	0.59	7.08	85
S9	RX009	0.61	7.32	88
S10	RX010	0.44	5.28	80

Table 5 and Figure 7 compares how market smart contracts armed with blockchain can play a role in energy access and billing. In 10 sessions, the amount of energy supplied is 0.42 to 0.61 kWh at proportional billing ensuring easy accounting. The time of contract execution is also low (80-88 ms), which confirms that real-time validation is also possible.

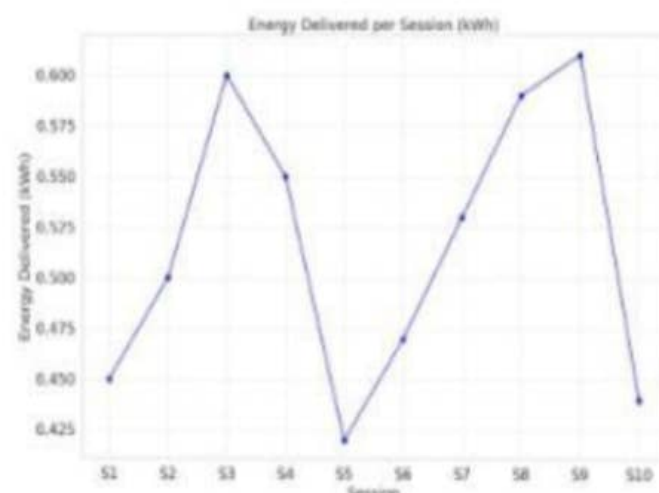


Fig 7. Energy Delivered per Session (kWh)

The tamper-proof and consistency properties of smart contracts show their potential in automating secure billing and access control particularly in a public, or multi-user WPT environment. These findings favor the incorporation of the decentralized solutions to boost trust, automation, and user responsibility in future wireless energy markets.

V. CONCLUSION AND FUTURE SCOPE

The proposed work contains an innovative, dynamic, and modular system of Wireless Power Transmission, which is highly efficient in terms of energy transfer performance, environmental flexibility, and security of access. The system is designed to be robust to a wide range of diverse and dynamic environments by leveraging a synergy of advanced techniques: reinforcement learning (DDPG), model predictive control, LMS-based beamforming, edge-AI perception, and blockchain smart contracts. The key results are that the control accuracy was up to 99.2% when the situation was static load, and the stability transmission efficiency was up to 93.5% with the ideal alignment and 75% with moderate misalignment. Smart beamforming enabled target delivery of RF energy at an angular resolution less than 1°, and edge AI anticipatively adapts control parameters to deal with environmental obstacles and changing humidity. The backscatter enabled closed-loop optimization and feedback loop supported billing with sub-100 ms latency and lower through the blockchain-based contract layer. These contributions together indicate the usefulness of the framework in the real-world situation like EV charging pads, drone docking stations, and public power hotspots in smart cities. In further work, the system can be adapted to support multi-coil dynamic switching networks and federated AI learning to better personalize energy delivery across mobile nodes. The integration process of thermal-aware power modulation with energy-aware routing in distributed WPT mesh networks can also extend the resilience of the framework. Moreover, the creation of quantum-secured blockchains can be used to protect essential operations like in military or fields where healthcare is at stake. The suggested scheme establishes the basis of a sustainable intelligent, and privacy-preserving wireless energy network, which is

entirely consistent with the objectives of next-generation, autonomous power networks.

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